

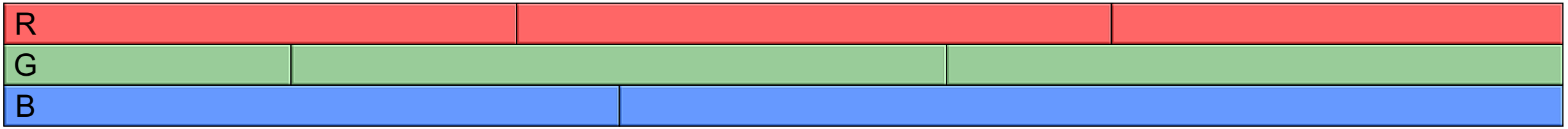
Dmitry Muravyev

Delta encoding

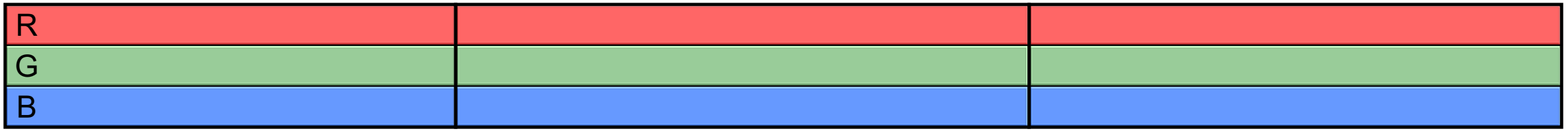
Sprint 3: Channels and optimization

Multi-channel compression

- Independent R/G/B channels compression:



- Joint RGB compression into a common block:

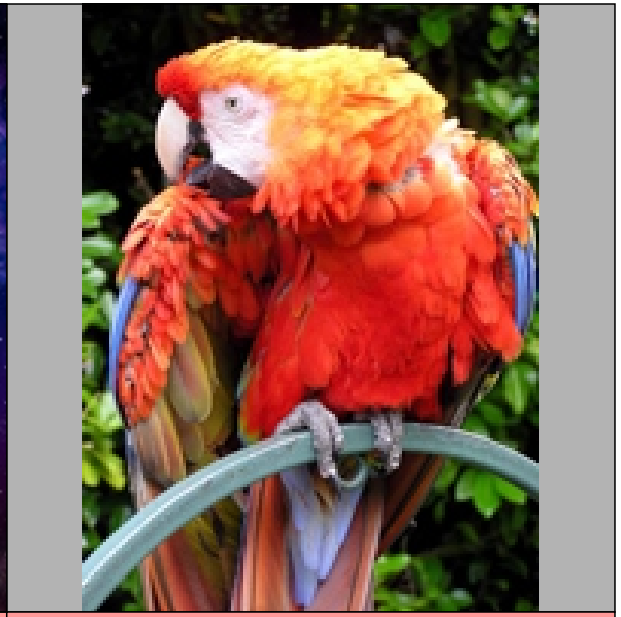


Multi-channel compression

RGB 888







9b, 1480■, 91.80%	7b, 3501■, 82.13%	5b, 3837■, 28.46%	5b, 2293■, 19.49%
7b, 3351■, 80.27%	6b, 4166■, 79.19%	5b, 3788■, 32.08%	5b, 2226■, 17.33%
7b, 3297■, 78.60%	6b, 3602■, 82.89%	5b, 3795■, 28.95%	5b, 2245■, 20.73%
8128■, 83.55%	11269■, 81.40%	11420■, 29.83%	6764■, 19.18%
8b, 3234■, 84.53%	7b, 4366■, 83.00%	5b, 4712■, 30.55%	5b, 3034■, 19.65%

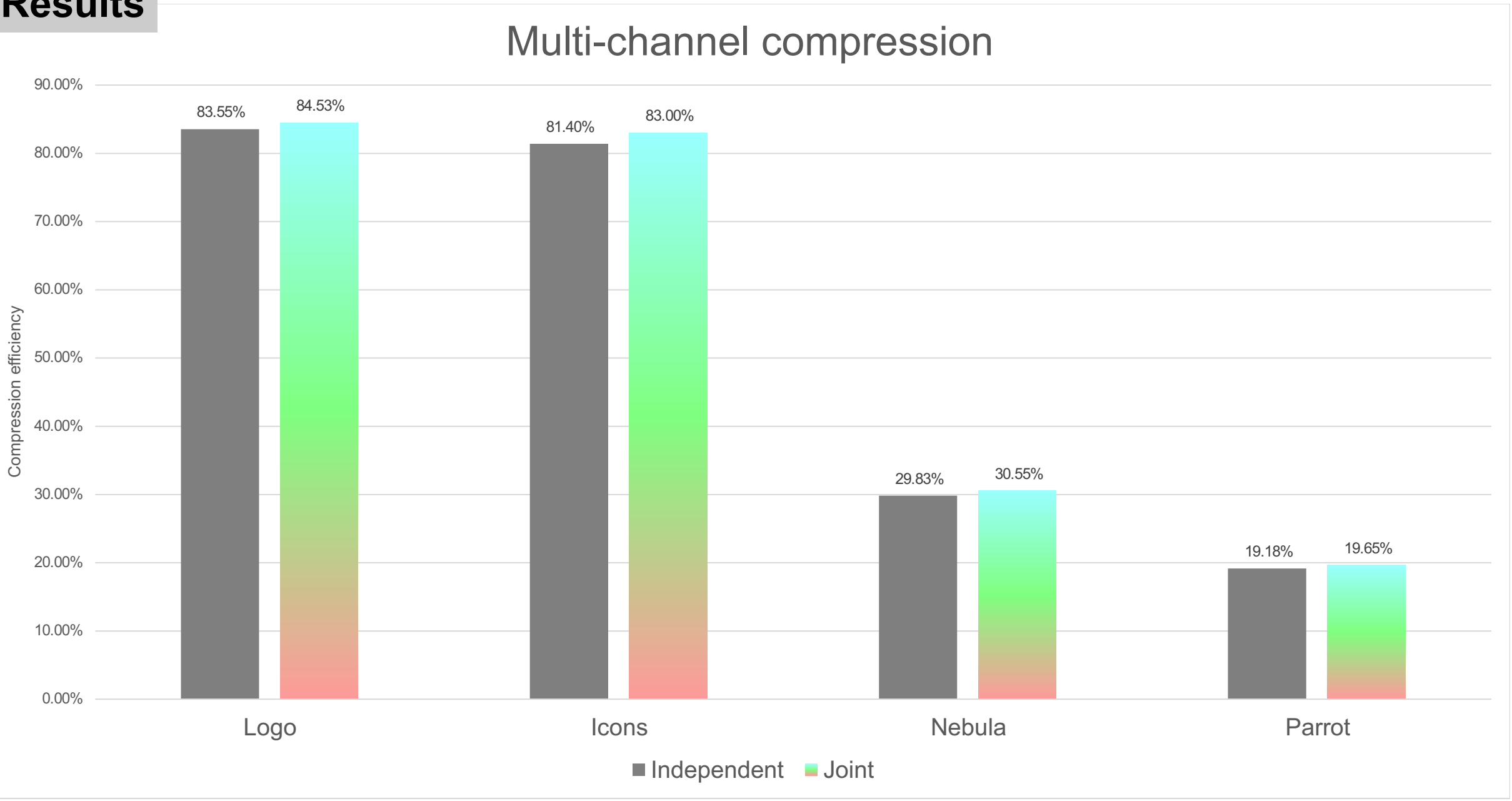
b – block size field (bits),

■ – resulting number of blocks,

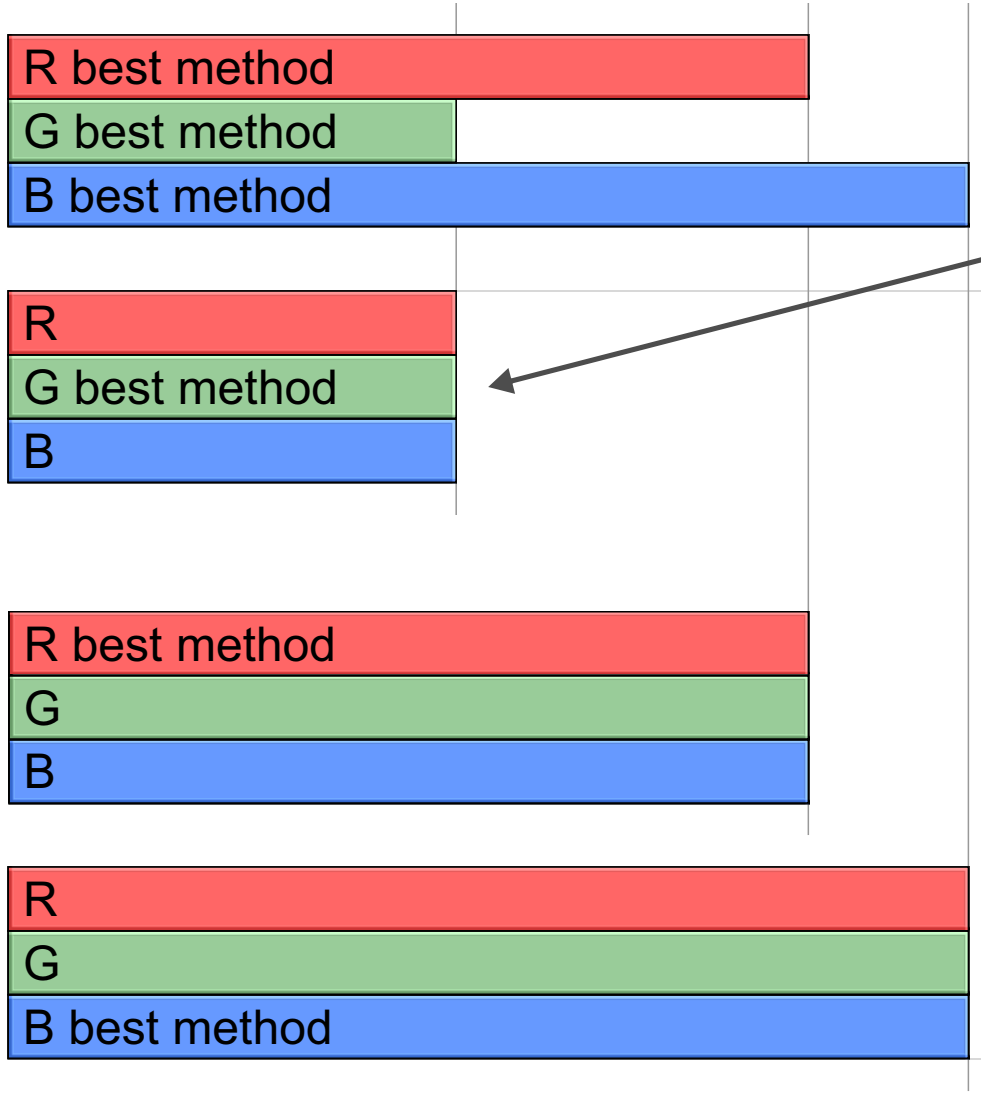
% – compression percentage (more is better)

Independent R/G/B channels compression
Overall result of independent compression
Joint RGB compression

Results


Multi-channel compression
Compression efficiency
90.00%
80.00%
70.00%
60.00%
50.00%
40.00%
30.00%
20.00%
10.00%
0.00%
83.55%
84.53%
81.40%
83.00%
29.83%
30.55%
19.18%
19.65%
Logo
Icons
Nebula
Parrot
Independent
Joint

Finding the best common block



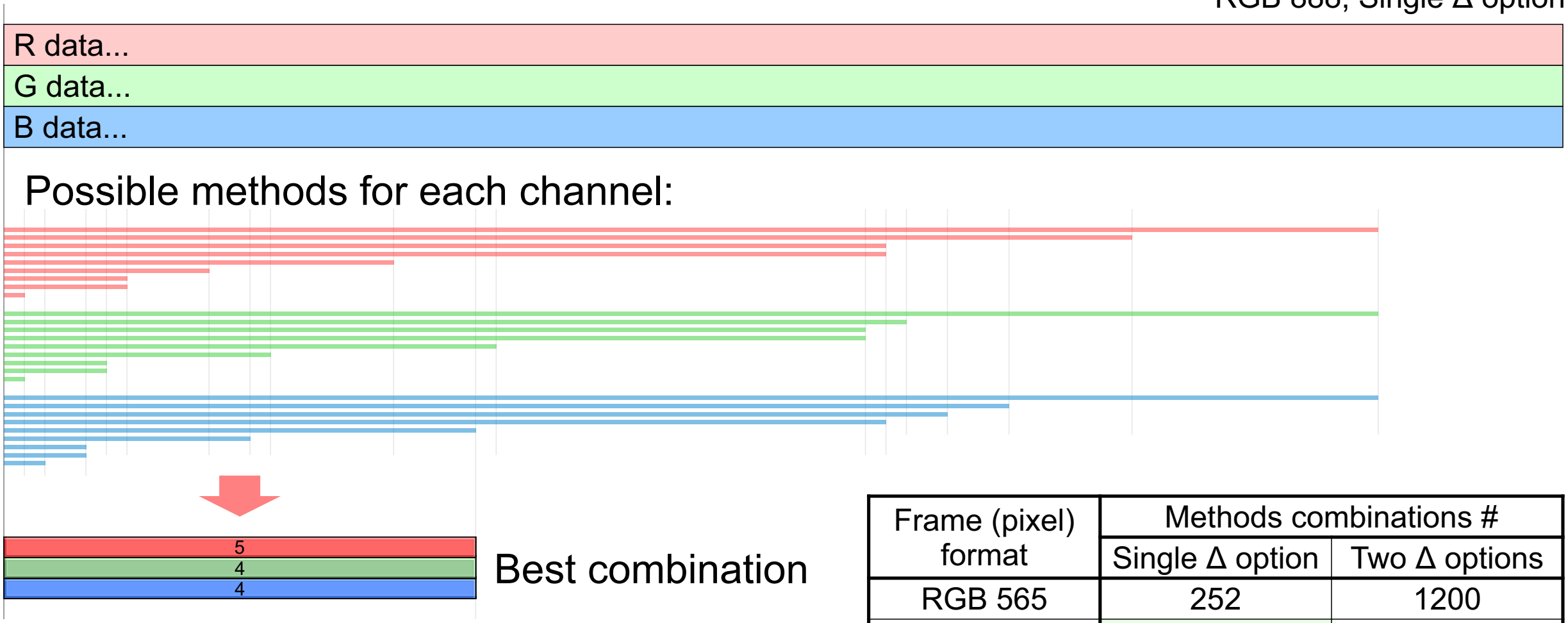
Trimming to the shortest method is not the best solution 👎

Much better is to choose the best block from the 3 options 👍

Fast way

Finding the best common block

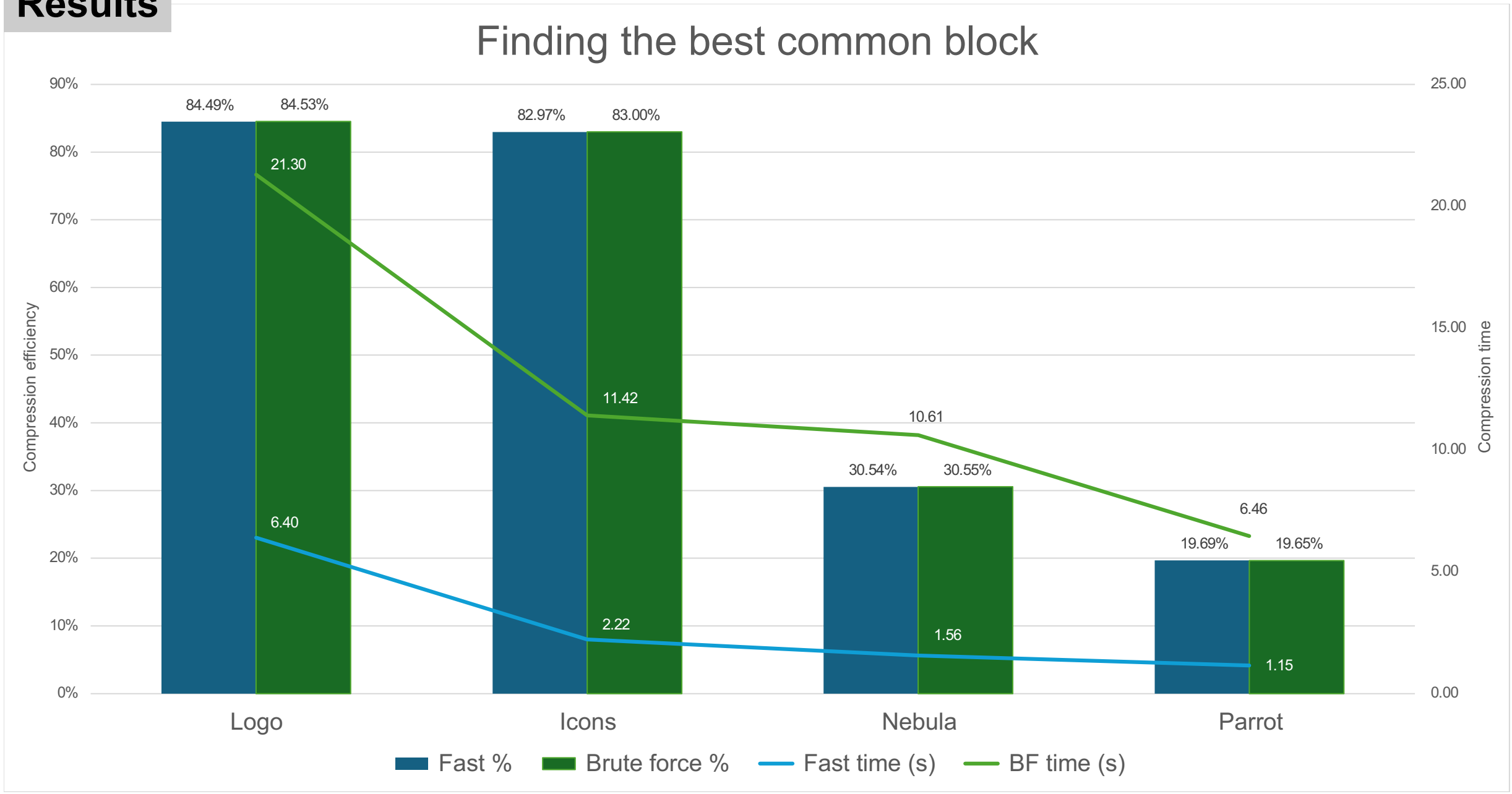
RGB 888, Single Δ option



Frame (pixel) format	Methods combinations #	
	Single Δ option	Two Δ options
RGB 565	252	1200
RGB 888	729	4096
RGB 48 bit	4913	32768
4ch × 32 bit	1 185 921	16 777 216

Brute force way

Results

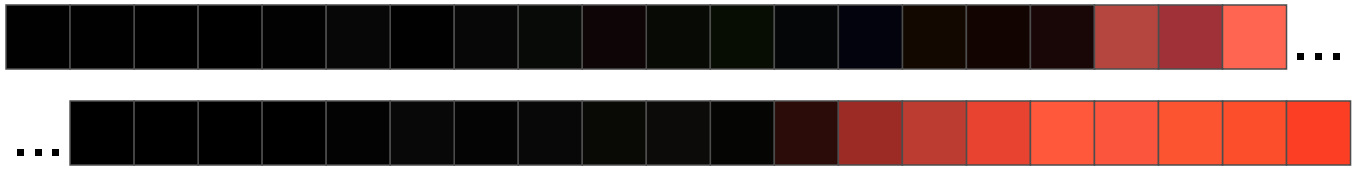

Finding the best common block
Compression efficiency
Compression time
90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
25.00
20.00
15.00
10.00
5.00
0.00
84.49%
84.53%
82.97%
83.00%
30.54%
30.55%
19.69%
19.65%
21.30
11.42
10.61
6.46
6.40
2.22
1.56
1.15
Logo
Icons
Nebula
Parrot
Fast %
Brute force %
Fast time (s)
BF time (s)

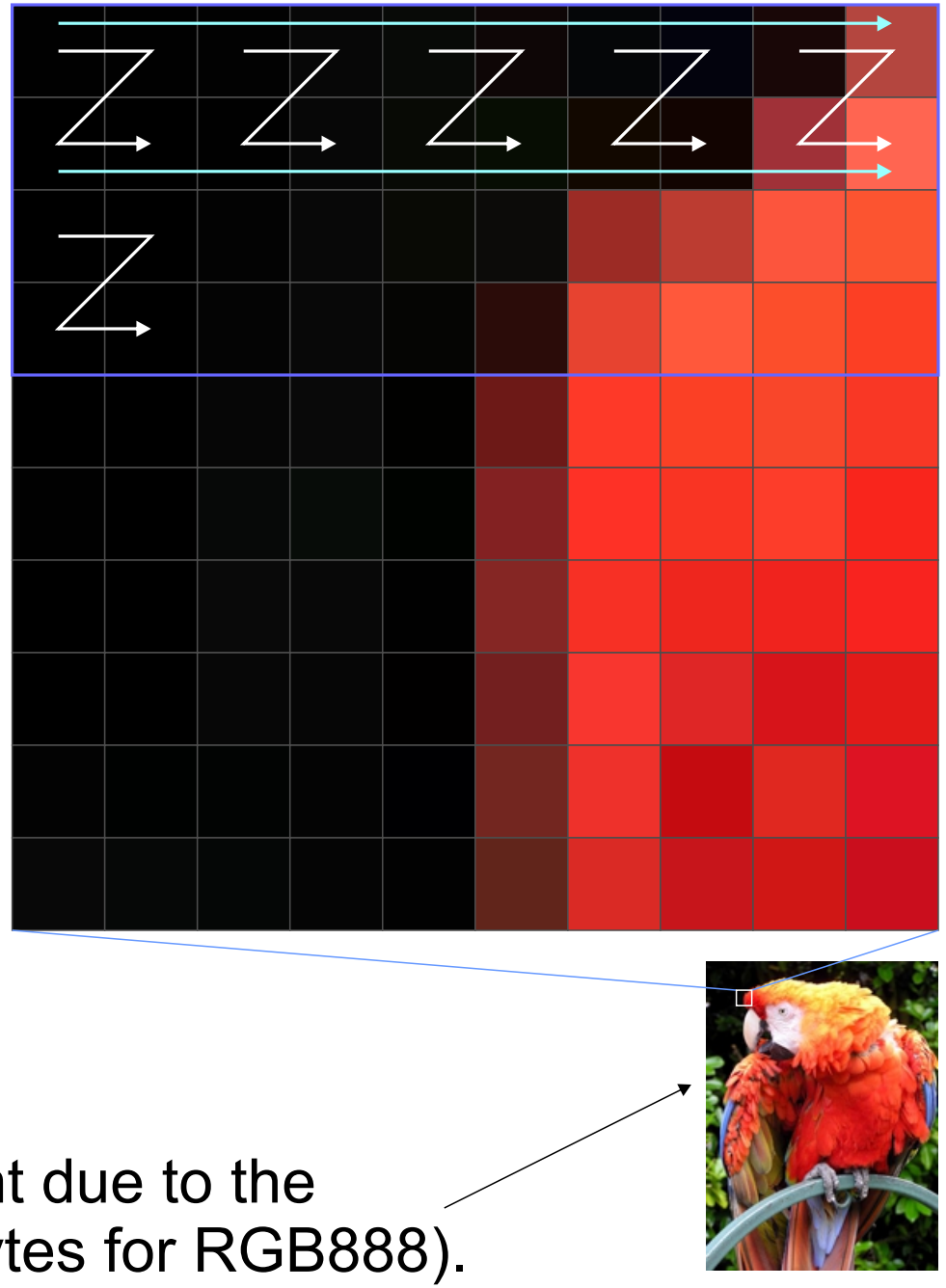
Splitting into squares

- Linear sequential pixel packing:



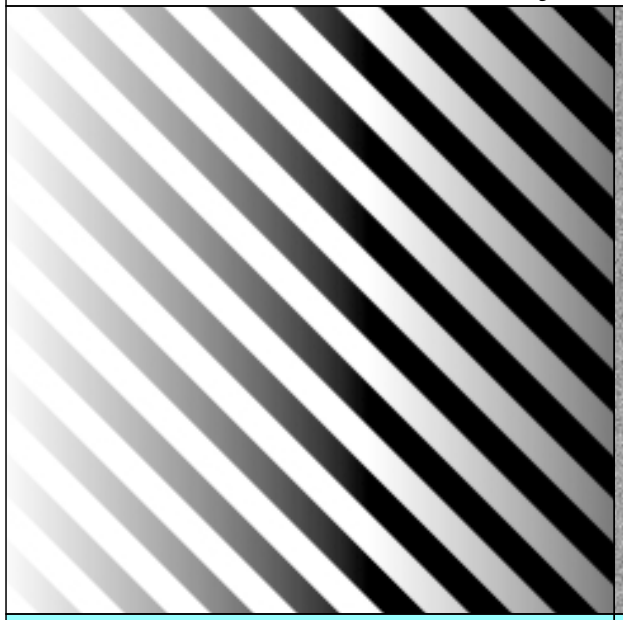
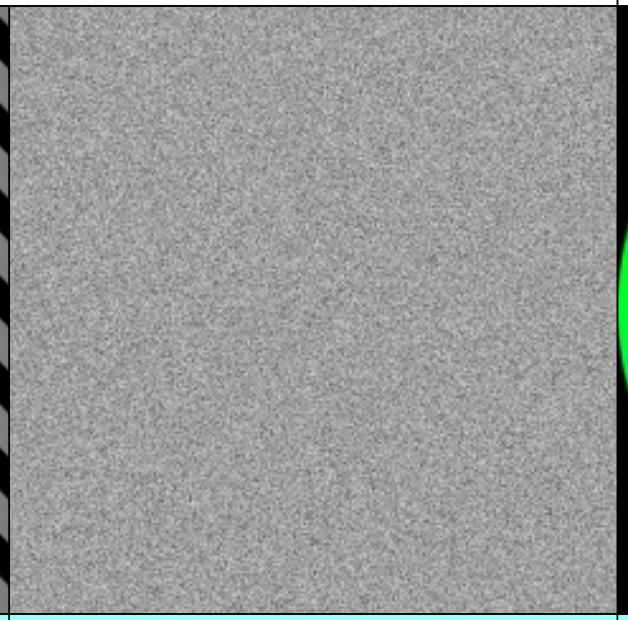
- Packing pixels with 2x2 squares:





* For this picture, the compression improvement due to the splitting was 1.91 percentage points (1722 bytes for RGB888).

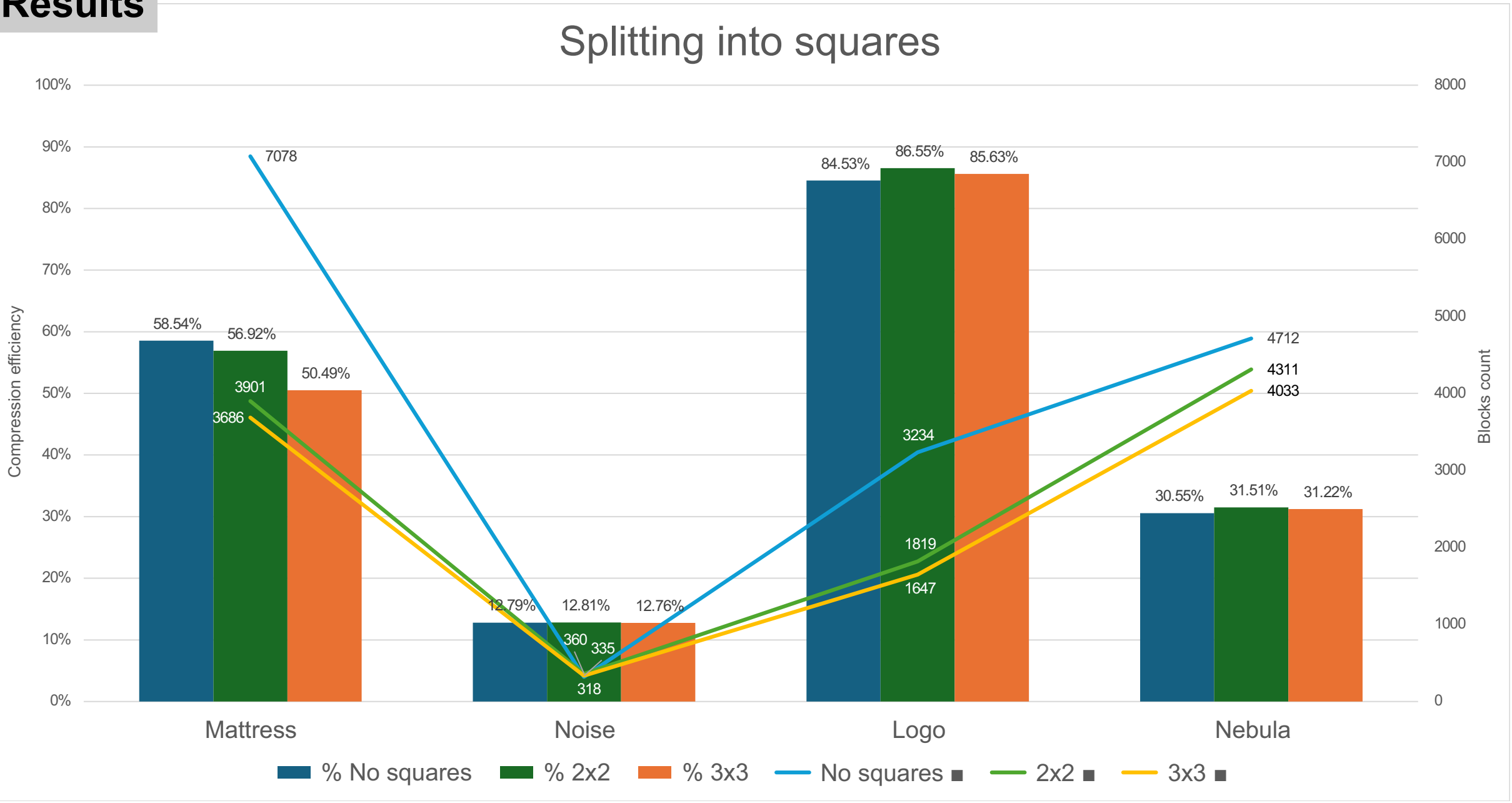
Splitting into squares

Grayscale 8 bit		RGB 888	
			
4b, 7078■, 58.54%	10b, 318■, 12.79%	8b, 3234■, 84.53%	5b, 4712■, 30.55%
5b, 3901■, 56.92%	10b, 360■, 12.81%	9b, 1819■, 86.55%	5b, 4311■, 31.51%
6b, 3686■, 50.49%	10b, 335■, 12.76%	9b, 1647■, 85.63%	5b, 4033■, 31.22%

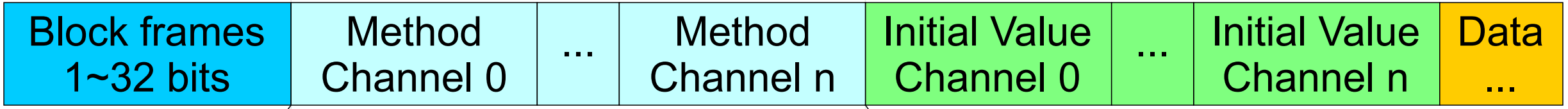
b – block size field (bits),
■ – resulting number of blocks,
% – compression percentage (more is better)

w/o splitting
Splitting into 2x2 squares
Splitting into 3x3 squares

Results

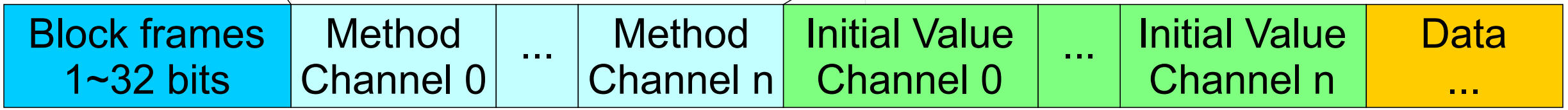

Splitting into squares
Compression efficiency
Blocks count
100%
90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
8000
7000
6000
5000
4000
3000
2000
1000
0
58.54%
56.92%
50.49%
12.79%
12.81%
12.76%
84.53%
86.55%
85.63%
30.55%
31.51%
31.22%
7078
3901
3686
360
335
318
3234
1819
1647
4712
4311
4033
Mattress
Noise
Logo
Nebula
% No squares
% 2x2
% 3x3
No squares
2x2
3x3

Header optimization

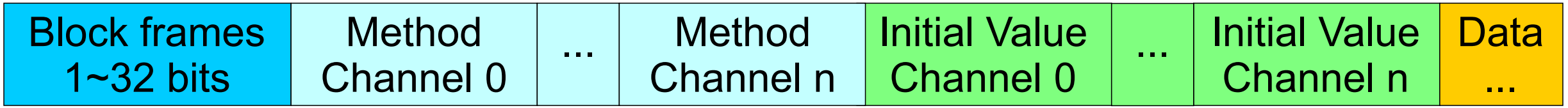


Channel bit depth	Bits per method declaration	
	Single Δ option	Two Δ options
2	2	2
3	2	3
4	3	3
5	3	4
6	3	4
7	3	4
8	4	4

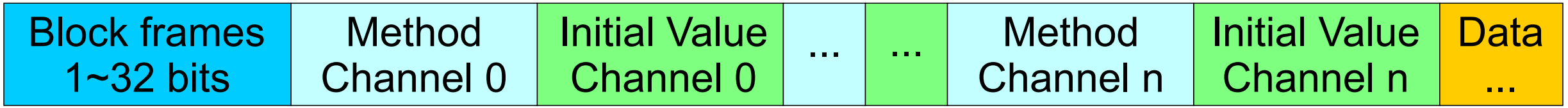
Reduce header size by using only one of the Delta calculation ways: either a Fixed window or an Adaptive Floating window.



Header optimization

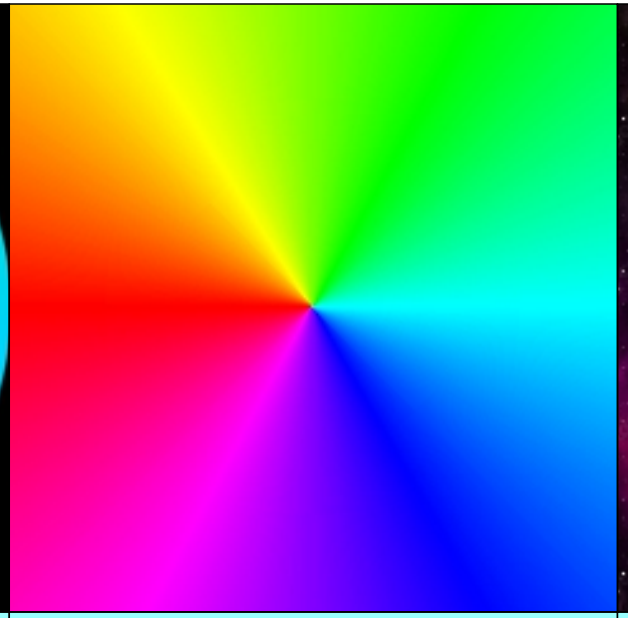
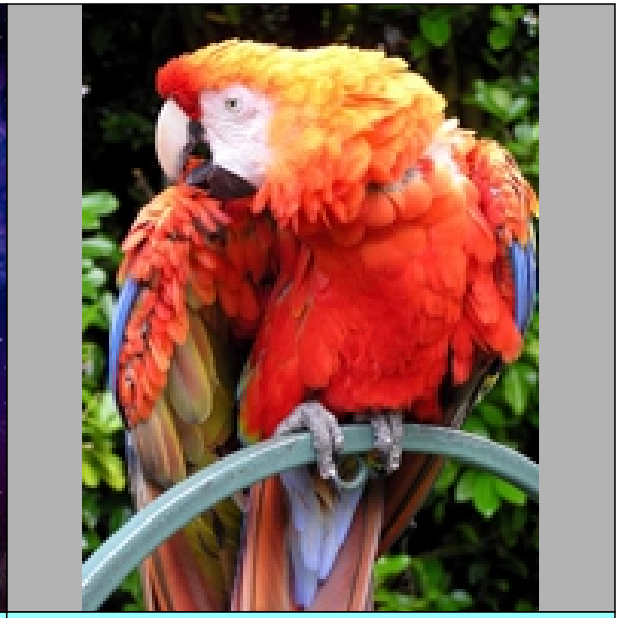


Reading Method declarations and Initial values in one loop can slightly speed up decompression.



Single Δ option

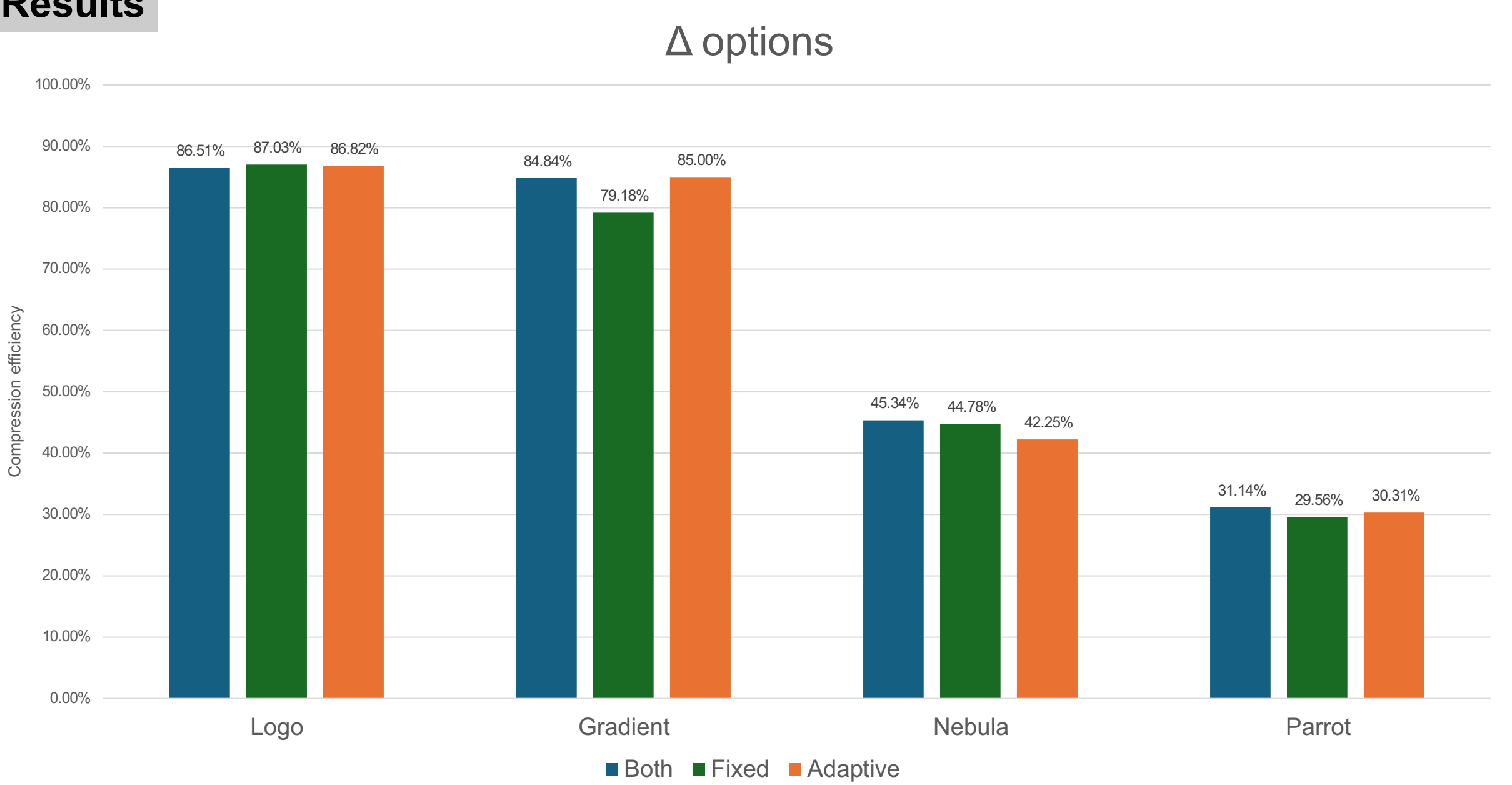
RGB 565, squares 2x2, BF

			
9b, 1882■, 86.51%	8b, 700■, 84.84%	5b, 4334■, 45.34%	5b, 2531■, 31.14%
9b, 1881■, 87.03%	5b, 2775■, 79.18%	5b, 4583■, 44.78%	5b, 2739■, 29.56%
8b, 2035■, 86.82%	8b, 685■, 85.00%	5b, 4661■, 42.25%	5b, 2774■, 30.31%

b – block size field (bits),
■ – resulting number of blocks,
% – compression percentage (more is better)

Both Δ options
Fixed window only
Adaptive floating window only

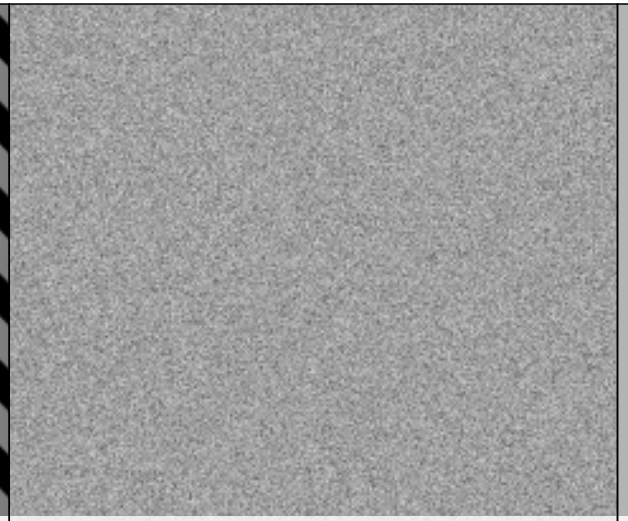
Results


Δ options
Compression efficiency
100.00%
90.00%
80.00%
70.00%
60.00%
50.00%
40.00%
30.00%
20.00%
10.00%
0.00%
86.51%
87.03%
86.82%
84.84%
79.18%
85.00%
45.34%
44.78%
42.25%
31.14%
29.56%
30.31%
Logo
Gradient
Nebula
Parrot
Both
Fixed
Adaptive

Final results



58.54%
Grayscale 8 bit



12.81%
Grayscale 8 bit



22.03%
Grayscale 8 bit

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13.24%
ASCII 8 bit



87.03%
RGB 565



45.34%
RGB 565



31.15%
RGB 565



45.70%
RGB 565

Thank you



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